

# Modal Triplet Theory: Foundations

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## Abstract

We develop a functional-analytic foundation for Modal Triplet Theory (MTT). The abstract architecture is a Hilbert bundle with three compatible vertical structures, a joint coherent spectral projector, a stabilization flow, and explicitly separate hypotheses for gap, invariance, existence, contraction, truncation, and admissibility. The canonical physical realization is a ten-dimensional bundle  $M_{10} \rightarrow Y_4$  with compact Riemannian fiber  $X_6$ ; the central circle is bundle data and is not counted as an additional product dimension. Strong commutation or a single total internal operator is assumed rather than inferred from notation. Complementary-mode stability uses a stable-semigroup estimate that remains valid for nonnormal generators. Projected time-step fixed points are distinguished from equilibria, and the existence, Lyapunov-promotion, and Banach gates are stated in self-contained form but imported from Fixed Points I, their canonical theorem source. Schur–Feshbach, projector-stability, and basin-robustness statements are given here with their required domains. Stabilization time, physical time, and renormalization scale are separated. Selection by reset is identified as a hybrid law unless derived from continuous upper dynamics. Lorentzian signature belongs to a hyperbolic principal symbol in a physical completion, not to a positive Hilbert-space Gram form. A complete admissibility ledger records the independent obligations inherited by every downstream MTT realization. A rank-three world-in-world comparison field and the selected q79 trace-split carrier are included as a typed geometry interface: their matching component counts do not by themselves derive a ten-dimensional manifold, Lorentzian spacetime, or a global intertwiner. The shared-circle claim is upgraded from fiberwise analogy to an exact finite differential-line theorem: one universal flat  $\mathbb{Z}_{64}$  line pulls back coherently to the q79 SpinC determinant, the  $1 + 2 + 3$  carrier, the root-plane complex structure, and the finite Reynolds Hessian. Boothby–Wang geometry independently identifies lens and Heisenberg nil manifolds as parallel curved prequantum circle bundles over different bases. These results are compatible but not identical, and neither compact circle flow is physical Lorentzian time. The curved connection compiler and nonlinear, support-stratified strain shadow now refine this interface without supplying a physical endpoint. Finite homotopy transfer is kept separate from the analytic fixed-point hypotheses and the selected physical action.

## Version 11 Revision Note

**Supersedes** Version 10; previous release records and revision notes are retained.

**Reason** Later owner results replace a proposed linear spectral-strain route and clarify which continuum data the analytic spine consumes.

**Resolution** Adds curved projective/Čech naturality, the nonlinear strain shadow and its support boundary, finite all-arity transfer, and the three-packet physical endpoint contract. Corrects the distinction between identity-coordinate orthogonality and general

polar variations, and states the real/complex comparison convention. Makes finite-rank spectral selection and the graph-norm inverse estimate explicit in their analytic statements.

**Retained** The scoped analytic theorems, mild-invariance correction and finite shared-line square; Fixed Points I–VI are not revised here.

**Open boundary** The physical endpoint, action, synthesis and compactification remain separate source obligations. This is an unreleased authoring revision, not a new release or physical status promotion.

## Version 10 Revision Note

**Supersedes** Version 9. The previous release remains public until a new release is approved.

**Reason** Condition algebraic tangency before promoting it to flow invariance.

**Resolution** Uses actual flow invariance and states sufficient mild-realization hypotheses for tangency.

**Retained result** Correctly scoped results and explanatory examples remain; no physical source-selection claim is promoted.

**Remaining boundary** Realization hypotheses and physical source or apparatus bridges remain separate obligations. This is an unreleased authoring revision.

## Revision note for version 9

**Supersedes.** *Modal Triplet Theory: Foundations*, version 8.

**Reason.** Version 8 contains the current analytic and shared-line results, but several dependencies were still easiest to read as a catalogue of formal statements. The release record also needed to distinguish the paper’s descriptive abstract from its correction history.

**Resolution.** Version 9 adds a reader map, plain-language explanations of the projector, damping, fixed-point, reduction, and geometry layers, and explicit ownership links to Fixed Points I and the Projection–Admissibility paper. It separates publication description from revision history and adds the managed reproducibility reference without changing a theorem tier.

**Retained result.** Every theorem, hypothesis, and open physical gate reported in version 8 is retained at the same logical strength.

**Remaining boundary.** The selected nonflat HYM endpoints, continuum intertwiner, physical action and Hilbert space, Lorentzian dynamics, and source-selected observables remain outside the Foundation theorem.

## Revision note for version 8

**Supersedes.** *Modal Triplet Theory: Foundations*, version 7.

**Reason.** Version 7 correctly typed the shared circle as line-bundle data, but it did not yet distinguish the curved Boothby–Wang Lens/Nil realizations from the later exact flat  $q_79$  differential-line theorem, nor did it record the resulting finite connection, holonomy, and Hessian identities.

**Resolution.** Version 8 retains the functional-analytic spine and adds the universal q79 flat differential line, its SpinC/CLN/root-plane/finite- Hessian pullbacks, the parallel Boothby–Wang realization theorem, and the conditional polarized-section readout. It also assigns the generic fixed-point gates to Fixed Points I and autonomous projection descent to the Projection–Admissibility paper; both are restated here as imported contracts rather than duplicated proofs.

**Retained result.** The coherent-projector and basin-local fixed-point architecture survives as a conditional control-and-reduction framework.

**Remaining boundary.** The flat finite theorem does not identify the nonzero-Chern physical HYM connection, prove the local strain-to-q79 continuum intertwiner, select a physical Hilbert space, or identify compact Reeb flow with time.

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## 1 How to read this foundation

The paper has two jobs. First, it supplies a reusable analytic language for selecting and controlling a coherent sector. Second, it shows where a physical MTT realization must add information that the abstract language cannot create.

The central picture is easier than the notation may initially suggest. At each point of a base space there is an internal space of possible modes. Vertical operators distinguish a selected low spectral sector from its complement. A stabilization flow acts on the full state. The Foundation asks whether the selected sector is mathematically well defined, whether unwanted modes decay, whether projected states exist and persist, and whether the discarded modes can be eliminated with a controlled error.

These are different questions. The existence of a spectral gap does not prove that the dynamics preserves the selected modes. The existence of a projected fixed point does not prove that the upper state is stationary. Uniqueness requires a stronger gate than existence. A successful internal reduction does not choose a Lorentzian spacetime, a probability rule, or a particle spectrum. Much of this paper is devoted to keeping those implications separate.

The geometry sections then specialize the language. They compare the local world-in-world strain carrier, the selected q79 trace carrier, the universal shared differential line, and the curved Boothby–Wang Lens and Nil realizations. These objects illuminate one another, but they are not identified merely because their dimensions or circle actions resemble one another.

A reader interested mainly in the analytic spine can move from the abstract Hilbert-bundle form to the joint projector, stable complementary dynamics, fixed-point gates, and reduction sections. A reader interested mainly in the current geometric interface can focus on the local comparison carrier, q79 interface, shared differential line, and Boothby–Wang realization before returning to the synthesis theorem. The formal statements record the exact hypotheses; the surrounding discussion explains why each statement is present and what it does not establish.

## 2 Status, scope, and logical vocabulary

This paper defines an abstract control and reduction architecture. It does not derive a particular quantum theory, gauge group, particle spectrum, spacetime equation, probability law, or numerical constant. Downstream claims must use one of the following statuses:

**Axiom or assumption.** Input structure of a realization.

**Conditional theorem.** A consequence proved from listed hypotheses.

**Characterization.** A necessary form for objects satisfying the premises.

**Reconstruction or embedding.** Recovery or representation of a known framework after compatible data are supplied.

**Calibration or postdiction.** Numerical agreement using target-related input or model selection.

**Held-out prediction.** A quantity not used in construction, calibration, scale choice, or branch selection.

**Interpretation.** A conceptual reading without theorem status.

The word “physical” is reserved for a realization equipped with a selected state space, local evolution law, observable map, and empirical interpretation.

This vocabulary is not decorative caution. It prevents a common failure mode in unification programs: a mathematically valid representation is first described as a reconstruction, then remembered later as a derivation. The status of a statement records what information entered before the conclusion was obtained. In particular, agreement with a datum used to select a branch is a profile check, not a held-out prediction.

### 3 Dimension-neutral architecture and physical realization

#### 3.1 Abstract Hilbert-bundle form

The abstract level deliberately says as little as possible about particles or spacetime. It provides a place where several internal mode structures can be compared over a common base. This is the level at which the fixed-point and reduction arguments are reusable: their validity should not depend on first assuming the physical conclusion one hopes to derive.

Let  $Y$  be a smooth base and let  $\mathcal{E} \rightarrow Y$  be a real or complex separable Hilbert bundle. A state belongs to a declared Sobolev space

$$\mathcal{H} = H^s(Y; \mathcal{E}),$$

with  $s$  chosen so that every nonlinear map and operator domain used below is well defined. The abstract results are dimension neutral.

The modal triplet is represented by three compatible vertical structures

$$(\mathcal{E}_i, A_i, P_i), \quad i = 1, 2, 3,$$

on the same internal Hilbert fiber. Here  $A_i$  is a nonnegative self-adjoint vertical operator and  $P_i$  is its selected low spectral projector. The triplet is not automatically a product of three coordinate manifolds.

The notation  $(\mathcal{E}_i, A_i, P_i)$  separates three roles. The bundle data say what kinds of internal states are available. The operator says how those states are spectrally organized. The projector says which cluster is retained in the proposed coherent description. Keeping the roles separate allows a later audit to determine whether a conclusion came from geometry, dynamics, or an explicit selection.

#### 3.2 Canonical physical realization

The canonical realization adds the physical arena used by the rest of the MTT program. It should be read as a specialization of the abstract language, not as a consequence of the word “triplet.” Four dimensions belong to the Lorentzian base and six to an internal Riemannian fiber; the two sides have different geometric jobs.

The canonical physical specialization is

$$\pi : M_{10} \longrightarrow Y_4, \quad X_x = \pi^{-1}(x), \quad \dim X_x = 6,$$

where  $Y_4$  is a four-dimensional globally hyperbolic Lorentzian base in the physical completion and each compact fiber  $X_x$  is Riemannian. In a local trivialization,  $M_{10} \simeq Y_4 \times X_6$ . Positive elliptic modal operators act vertically on  $X_6$  and do not create additional causal directions.

If  $X_6$  is explicitly factorized as  $F_1 \times F_2 \times F_3$ , then  $\sum_i \dim F_i = 6$ . A shared phase circle is represented by a principal  $U(1)$  bundle or Hermitian line bundle  $L_{\text{cen}} \rightarrow X_6$ . It is not appended as a seventh independent product coordinate. Recursive nil/lens/circle descriptions may be used as bundle or filtration data, but their dimensions must not be added unless an actual product decomposition is proved.

This corrects two tempting pictures. A fiber direction is not automatically an additional causal direction, and a common circle reused by several lanes is not a new coordinate every time it appears. The six-dimensional count belongs to the actual fiber. Circle, Lens, and Nil language can further organize that fiber only through explicitly supplied bundle maps.

### 3.3 Local comparison carrier and selected q79 interface

The next two carriers answer different questions. The world-in-world field describes a local comparison between two rank-three frames. The q79 carrier describes globally organized sheet data on a selected degree-three cover. Their common one-two-three profile is a clue that motivates an intertwiner; it is not itself that intertwiner.

Let  $TP$  and  $TI$  be oriented Euclidean rank-three bundles over a common base. A world-in-world comparison field is a typed section

$$Q_{\text{WW}} \in \Gamma(\text{Hom}(TP, TI)).$$

After choosing local frames,  $Q_{\text{WW}}$  has nine matrix components. This is not dimension multiplication: the total space of a rank-three vector bundle over a three-dimensional base has dimension  $3 + 3 = 6$ .

Polar decomposition gives three orientation coordinates and six strain coordinates. At the identity, their infinitesimal representatives have the orthogonal Frobenius split

$$\text{Mat}(3, \mathbb{R}) = \mathfrak{so}(3) \oplus \text{Sym}(3, \mathbb{R}), \quad 9 = 3 + 6.$$

At a general positive polar factor  $U$ , the variations  $\Omega U$  and  $\delta U$  need not be Frobenius-orthogonal. The component identity survives, but a metric in polar coordinates must be pulled back explicitly. Relative to a selected orthonormal flag, the strain space has the orthogonal decomposition

$$\text{Sym}(3, \mathbb{R}) = \mathbb{R}I_3 \oplus \mathcal{D}_0 \oplus \mathcal{O}, \quad \dim(\mathbb{R}I_3, \mathcal{D}_0, \mathcal{O}) = (1, 2, 3),$$

where  $\mathcal{D}_0$  is traceless diagonal and  $\mathcal{O}$  is symmetric off-diagonal. Thus

$$1 + 3 \times 3 = (1 + 3) + (1 + 2 + 3) = 4 + 6 = 10$$

is an exact component identity once one additional ordering scalar is supplied. It is not a tangent-space decomposition of  $M_{10}$ , does not select a  $3 + 1$  Lorentzian base, and does not globalize without transition data.

For the selected degree-three q79 spectral cover, let

$$\mathcal{A} = \pi_* \mathcal{O}_C, \quad \mathcal{A} = \mathcal{O} \oplus \mathcal{A}_0, \quad \text{rank}(\mathcal{O}, \mathcal{A}_0, \mathcal{A}) = (1, 2, 3),$$

where  $\mathcal{A}_0 = \ker \text{Tr}$ . The corresponding common carrier is

$$\mathcal{H}_{\text{CLN}} = L_{\text{shared}} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A}), \quad \text{rank } \mathcal{H}_{\text{CLN}} = 6.$$

This proves the selected complex rank filtration, not its identification with the real strain decomposition. A complex-linear comparison first complexifies the latter; a real comparison

instead needs a compatible real form of the former. A linear identification would then require a same-source bundle-and-connection intertwiner. The line bundle  $L_{\text{shared}}$  carries common  $U(1)$  phase or holonomy data; it is neither an extra product dimension nor physical Lorentzian time.

In plain terms, the local carrier says how a three-directional comparison can deform, while the q79 carrier says how three sheets split into common, trace-free, and full information. Both supply six organized directions. What is missing is a map showing that transporting, differentiating, and measuring one carrier is equivalent to doing the same with the other. Rank agreement alone cannot provide those dynamical statements.

### 3.4 Universal q79 differential line and finite operator square

The shared-line construction addresses one narrower part of that bridge. It asks whether the phase line used in several finite q79 sectors is literally the pullback of one differential object, with the same connection and holonomy, rather than a collection of merely isomorphic circles. This is important because matching fibers at one point would not control what happens around a loop.

The phrase “one shared circle” has a global meaning only after its classifying maps and connection are retained. Let  $\mathcal{L}_{64}^{\text{univ}} \rightarrow B_{\nabla} \mathbb{Z}_{64}$  be the universal flat Hermitian line associated to the primitive character

$$\chi_1(n) = \exp(2\pi i n/64).$$

On the unbranched q79 sheet stack, let  $c_{\text{sheet}} : B^\circ \rightarrow BS_3$  classify the sheet local system. Since  $S_3^{\text{ab}} \cong \mathbb{Z}_2$ , there is exactly one nontrivial homomorphism to  $\mathbb{Z}_{64}$ ,

$$h_{S_3}(\sigma) = 32 \epsilon(\sigma), \quad \epsilon(\sigma) = \begin{cases} 0, & \sigma \text{ even,} \\ 1, & \sigma \text{ odd.} \end{cases}$$

Put

$$c_{\text{sh}} = B(h_{S_3}) \circ c_{\text{sheet}}, \quad (L_{\text{sh}}, \nabla_{\text{sh}}) = c_{\text{sh}}^*(\mathcal{L}_{64}^{\text{univ}}, \nabla_{64}).$$

For both admissible odd roots  $r \in \{1, 33\}$ ,

$$\chi_r \circ h_{S_3} = \text{sgn}.$$

Thus the associated determinant sign line and  $L_{\text{sh}}$  are canonically parallel-isomorphic: their connections and every loop holonomy agree, not merely their fiber dimensions.

**Theorem 3.1** (Finite shared-line and Hessian intertwiner). *On the selected q79 flat root-stack symbol, the same pullback  $L_{\text{sh}}$  tensors all three trace lanes*

$$\mathcal{H}_{\text{CLN}} = L_{\text{sh}} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A}).$$

*Its scalar holonomy commutes with the trace, trace-zero, and full projectors, with the root-plane quarter-turn  $J_{DE}$ , and with the normalized finite Reynolds operators*

$$P_{\text{Haar}} = \frac{1}{6} \sum_{g \in S_3} \rho(g), \quad H_{\text{fin}} = \kappa_{\text{fin}}(I - P_{\text{Haar}}).$$

*For the selected two-copy sheet/edge representation,*

$$\text{rank } P_{\text{Haar}} = 2, \quad \text{spec}(H_{\text{fin}}/\kappa_{\text{fin}}) = \{0^{\times 2}, 1^{\times 4}\}, \quad H_{\text{TT}} = \kappa_{\text{fin}} I_2.$$

*The lifted Hessian is exactly  $I_{L_{\text{sh}}} \otimes H_{\text{fin}}$  and introduces no dimensionless fit.*

*Proof.* The character identity gives the parallel determinant-line isomorphism. Scalar line holonomy commutes with every finite sheet operator. Haar averaging is an orthogonal projector, so  $I - P_{\text{Haar}}$  is the complementary projector. Direct decomposition of the selected two-copy permutation representation gives the displayed ranks and spectrum. Tensoring by the same line preserves these identities and proves the intertwining statement.  $\square$

The practical content of the theorem is that the same finite phase transport acts compatibly on all three q79 lanes and on the finite Hessian. No dimensionless coefficient is adjusted to make the spectra agree. This closes a finite same-source square that earlier versions of the theory described only by analogy.

The theorem is exact at flat differential-character and finite-symbol tier. It does not identify  $\nabla_{\text{sh}}$  with a nonflat physical HYM connection: a flat trace-free carrier has vanishing real characteristic curvature, whereas the active physical bundle has nonzero  $c_2$ . The continuum target is a spectral-symbol functor and a unitary parallel Hessian comparison, not literal equality of those connections.

### 3.5 Boothby–Wang Lens and Nil realizations

The Boothby–Wang construction explains why Lens and Nil repeatedly appear beside a circle in the MTT corpus. Both can be built as curved circle bundles, but they sit over different bases and carry different curvature. The common idea is prequantum circle transport; the resulting total spaces are not nested copies of one another.

Let  $(B, \omega)$  be an integral symplectic manifold. The Boothby–Wang construction produces a principal circle bundle

$$U(1) \longrightarrow P_k \xrightarrow{\pi} B, \quad c_1(P_k) = k[\omega],$$

with connection/contact form  $\alpha_k$  satisfying, in the chosen normalization,

$$d\alpha_k = 2\pi k \pi^* \omega.$$

For  $B = \mathbb{CP}^1$  this total space is the lens space  $L(k, 1)$ ; hence  $k = 3$  gives  $L(3, 1)$  exactly at bundle-topology level. For an integral area form on  $T^2$ , the total space is a quotient of the Heisenberg group by a lattice and is a three-dimensional contact nilmanifold [17, 18].

Consequently Lens and Nil are parallel realizations of one prequantization schema over different bases and curvature classes. They are not literally nested manifolds. Nor are they one literal line bundle: the rigorous common target is the differential classifier  $BU(1)_{\nabla}$  with separate maps

$$c_{\text{Lens}} : \mathbb{CP}^1 \rightarrow BU(1)_{\nabla}, \quad c_{\text{Nil}} : T^2 \rightarrow BU(1)_{\nabla}.$$

The flat q79 classifying map factors through  $B_{\nabla} \mathbb{Z}_{64} \rightarrow BU(1)_{\nabla}$ . Because the Boothby–Wang connections above have nonzero curvature while the q79 root-stack line is flat on  $B^\circ$ , the constructions cannot be identified connection by connection. A stronger MTT “same circle” theorem would have to select the three classifying maps and coherent comparison 2-cells on a declared correspondence space.

After a polarization, weight- $m$  equivariant functions on  $P_k$  correspond to sections of the associated line  $L^m$ . In a positive holomorphic polarization this gives the familiar readout

$$\mathcal{H}_m = H^0(B, L^m).$$

This is a conditional geometric-quantization Hilbert space, not a Hilbert space selected by the abstract MTT axioms. The periodic Reeb flow is the vertical phase action and must not be identified with noncompact physical time [19].

These two circle constructions therefore complement rather than replace one another. The flat q79 line provides an exact finite holonomy and Hessian intertwiner. The curved Boothby–Wang bundles explain how Lens and Nil geometries can arise from a common construction principle. A future continuum theorem must relate them on a declared correspondence space while preserving their different curvature data.

### 3.6 The current curved and nonlinear interface

The QM owner makes the continuum side explicit [1, 3, 4]. Smooth finite matrix projectors describe bundles over a base with infinitely many possible modes; they do not flatten nonzero characteristic curvature. The Čech compiler represents a supplied connection by  $(p, \Gamma)$ , not by the Grassmann projector connection alone. Unitary naturality transports the metric, connection, operator domains and spectral windows together; raw fixed Fourier windows need not reduce the physical operator. CLN is local, auxiliary or filtrational structure, not the global q79 topology. In particular, this paper does not identify that topology with a literal  $S^1 \times \text{Lens} \times \text{Nil}$  product or put a parallel flag inside an irreducible physical HYM bundle.

The same owner excludes the old phase-stable linear rank-six root-space route by its incompatible  $S_3$  character. The replacement is the nonlinear spectral-strain map

$$\Phi(X) = (d_0, d_1, d_2, |z_{12}|^2, |z_{20}|^2, |z_{01}|^2) \quad (X \in \text{Herm}(3)).$$

It forgets the triangle phase; the full generic relative-phase quotient is seven-dimensional. On the regular image, the derivative  $J = D\Phi$  and a supplied positive source Hessian’s reduced Green  $G_Q$  yield the shorted Hessian  $(JG_QJ^*)^{-1}$  where the reduced covariance is invertible, by minimizing over the discarded directions, not by a bundle isometry [5]. For a TT-only observable, project  $J$  before shorting. The later global map has eight edge-support strata: a zero edge has a quadratic normal jet and a degree-one normal strain cost, not an ordinary six-dimensional boundary Hessian [6]. Rank-six surjectivity, or the required local  $C_4$  derivative  $J_{DE}$ , forces the regular stratum. This retires regularity as an independent input under that stronger contract without selecting the physical endpoint, Green operator or same-source  $S_3 \times C_4$  lift. The finite quarter-turn is a tangent operation, not a global automorphism of the strain orthant.

## 4 Three independent evolution parameters

Several earlier MTT arguments used the word “flow” for physically different operations. This section prevents that ambiguity. A mathematical relaxation can tell us how an iteration approaches a stable sector without being the time experienced by a physical observer. Renormalization, in turn, compares descriptions at different scales and need not be either process.

The foundation distinguishes:

1. a stabilization flow  $R_\tau$  with control parameter  $\tau \geq 0$ ;
2. a physical propagator  $U(t_2, t_1)$ , supplied only by a selected physical completion; and
3. a renormalization or coarse-graining scale  $\mu$ .

No equality among  $\tau$ ,  $t$ , and  $\log \mu$  is assumed. A theorem relating any two of them must specify the map, units, domain, and approximation error.

For example, a heat-flow parameter has dimensions fixed by the operator that generates it. A physical time coordinate belongs to a Lorentzian initial-value problem. Writing both with

an ordered symbol does not make their units or causal meanings agree. The same warning applies to a phase angle on the shared circle: a clock may use repeated phase to measure elapsed time, but phase is not time itself.

We write the stabilization equation as

$$\partial_\tau \Psi = F(\Psi), \quad R_\tau(\Psi_0) = \Psi(\tau).$$

Well-posedness is imposed on a declared interval and invariant domain; global well-posedness is not part of the abstract foundation unless proved in a specific realization.

## 5 Joint vertical operator and coherent projector

The coherent projector is the mathematical version of asking which internal modes pass all selected spectral tests. If the tests are compatible, their low-mode projectors can be intersected without depending on the order in which they are applied. If they are not compatible, multiplying the symbols  $P_1 P_2 P_3$  need not produce an orthogonal projector at all.

**Assumption 5.1** (Joint spectral structure). For each base point, the  $A_i$  are nonnegative self-adjoint operators whose spectral measures strongly commute. Their quadratic forms have a common dense domain. Equivalently, a realization may provide one nonnegative self-adjoint total internal operator  $A_{\text{int}}$  directly.

Under strong commutation define

$$P = \prod_{i=1}^3 \mathbf{1}_{I_i}(A_i), \quad Q = I - P,$$

for declared isolated low spectral sets  $I_i$ . The product is then an orthogonal projector independent of ordering. Base-only coefficients or separate variable names do not, by themselves, prove strong commutation.

A finite-dimensional toy model makes the issue visible. Three diagonal Hermitian matrices in one basis have commuting spectral projectors, and the joint low sector is the span of basis vectors accepted by all three. Rotate one matrix into a noncommuting basis and successive projections can depend on order. The abstract assumption is the infinite-dimensional replacement for the common-basis situation. A realization that cannot prove it should define one total operator instead of treating the product as canonical.

For harmonic selection one may instead use the form sum  $A_{\text{int}} = A_1 + A_2 + A_3$ . Nonnegativity gives

$$\ker A_{\text{int}} = \bigcap_i \ker A_i,$$

provided the form sum is well defined. The coherent projector is then the spectral projector of  $A_{\text{int}}$  at zero.

**Assumption 5.2** (Internal gap and Sobolev boundedness). There is  $\lambda_{\text{int}} > 0$  such that, in quadratic-form sense,

$$A_{\text{int}} \succeq \lambda_{\text{int}} Q,$$

and  $P, Q$  extend boundedly to every Sobolev space used by the dynamics.

The internal gap separates the selected fiber cluster from complementary fiber modes. It does not by itself imply invariance under  $R_\tau$ , existence of a fixed point, coherent contraction, suppression of arbitrarily high four-dimensional energy, or selection of a physical state.

The gap is therefore a separation statement, not a complete stability theorem. It says that leaving the selected internal cluster costs a positive amount in the vertical operator. Whether the actual dynamics respects or damps that separation is the subject of the next section.

## 6 Stable complementary dynamics

The complementary sector contains the modes removed from the coherent description. For a controlled reduction, small disturbances in this sector must not grow without bound before they are eliminated. This is a dynamical requirement, stronger than the spectral gap by itself.

Let  $\Psi_*$  be a reference state at which the stabilization vector field is Fréchet differentiable, and set  $L = DF(\Psi_*)$ . The stable sign convention is that  $L_{QQ} = QLQ$  generates decay:

$$\|e^{\tau L_{QQ}}\| \leq M_Q e^{-\omega_Q \tau}, \quad M_Q \geq 1, \quad \omega_Q > 0.$$

This estimate, rather than spectral abscissa alone, is authoritative for a nonnormal generator.

**Proposition 6.1** (Gap-to-decay under a bounded perturbation). *Suppose  $Q\mathcal{H}$  is invariant and*

$$L_{QQ} = -\kappa A_{\text{int}}|_{Q\mathcal{H}} + B_Q, \quad \kappa > 0, \quad B_Q \in \mathcal{B}(Q\mathcal{H}).$$

*If  $\omega_Q := \kappa \lambda_{\text{int}} - \|B_Q\| > 0$ , then*

$$\|e^{\tau L_{QQ}}\| \leq e^{-\omega_Q \tau}.$$

*Proof.* The self-adjoint semigroup generated by  $-\kappa A_{\text{int}}|_{Q\mathcal{H}}$  has norm at most  $e^{-\kappa \lambda_{\text{int}} \tau}$ . The bounded perturbation estimate gives the additional factor  $e^{\|B_Q\| \tau}$ .  $\square$

The proposition describes the simplest transparent mechanism. The positive internal operator supplies decay after the required minus sign is included, while the bounded perturbation spends part of that decay margin. Stability survives only when the remaining margin  $\omega_Q$  is positive.

For a general sectorial or nonnormal  $L_{QQ}$ , the constants  $M_Q$  and  $\omega_Q$  must be proved directly. A positive vertical operator cannot be identified with the stabilization generator without the minus sign and the lower-order terms.

Nonnormality matters because eigenvalues alone can hide large transient growth. The prefactor  $M_Q$  records that possibility. Downstream calculations that quote only a negative spectral abscissa have not yet supplied the semigroup control required by this Foundation.

## 7 Independent logical gates

The following list is the paper’s main dependency map. It replaces the older idea that one favorable “coherence” inequality automatically supplies the entire physical conclusion. Each gate answers a separate failure question: is the sector defined, is it preserved, does a returning state exist, is that state truly stationary, is it unique, and is the reduction accurate?

The following gates are independent and must not be collapsed:

**Gap.** Spectral separation for  $A_{\text{int}}$ .

**Projector stability.** Persistence and regularity of  $P$  under parameter or curvature variation.

**Invariance.**  $R_t(P\mathcal{H}) \subseteq P\mathcal{H}$  for every time in the stated existence interval. Algebraic tangency  $QF(Pu) = 0$  alone is not a substitute for this flow statement.

**Existence.** A fixed-point theorem applies on a declared invariant domain.

**Equilibrium identification.** A projected time-step fixed point is shown to be stationary.

**Contraction.** The coherent reduced map is strictly contractive.

**Truncation.** The influence of  $Q$  on  $P$  is quantitatively bounded.

**Selection.** A continuation or reset rule is supplied at admissibility exit.

No gate in this list follows solely from the gate preceding it.

For a mild equation with a commuting linear semigroup and a globally Hilbert-space-Lipschitz nonlinear term, tangency of that nonlinear term to  $P\mathcal{H}$  is sufficient: the restricted mild solution stays in the subspace and uniqueness identifies it with the full solution. This is the mild-invariance criterion imported from Fixed Points I [12]. In a different Sobolev phase space, a locally defined flow, or a weak realization, continuation, domain compatibility, and uniqueness must be proved there before the same inference is used.

A simple projected movie illustrates the distinction. Suppose the visible frame after one step looks exactly as it did before. The hidden part of the state may nevertheless have moved and then been erased by projection. That is a projected time-step fixed point, not an equilibrium of the full dynamics. Likewise, two different upper states may cast the same returning visible frame, so existence does not imply uniqueness.

## 8 Imported fixed-point gates

The Foundation needs fixed-point results in order to state its complete control contract, but Fixed Points I is their canonical owner. They are reproduced here as readable hypotheses and conclusions, without duplicating the owning proofs. The three routes answer increasingly strong questions: existence of at least one returning state, promotion of return to stationarity, and uniqueness with convergence.

Let  $K \subset P\mathcal{H}$  be nonempty, closed, bounded, and convex and define

$$T_\tau = PR_\tau|_K.$$

Fixed Points I is the canonical theorem source for the following three gates [12]. They are restated here, with all hypotheses and conclusions, so that the Foundation remains independently readable.

**Existence gate.** Assume  $T_\tau(K) \subset K$  and either

1.  $T_\tau : K \rightarrow K$  is continuous and compact; or
2.  $T_\tau$  is continuous and condensing for a declared measure of noncompactness.

Then there is  $u_* \in K$  with  $T_\tau u_* = u_*$ . This is the Schauder or Darbo–Sadovskii gate, respectively.

The conclusion is a *projected time-step fixed point*. It is not named an equilibrium merely because  $PR_\tau u_* = u_*$ .

Compactness and condensing behavior are existence mechanisms. They can bring an infinite-dimensional image into a sufficiently controlled set for a fixed point theorem to apply. Neither mechanism says that every initial state converges to the same point.

**Equilibrium-promotion gate.** Suppose  $P\mathcal{H}$  is invariant, so that  $PR_\tau u_* = R_\tau u_* = u_*$ , and along the orbit

$$\mathcal{C}(R_\tau u) - \mathcal{C}(u) = - \int_0^\tau \mathcal{D}(R_s u) ds, \quad \mathcal{D} \geq 0,$$

where  $\mathcal{D}(v) = 0$  exactly when  $F(v) = 0$ . Then the fixed point from the existence gate is an equilibrium: endpoint recurrence makes the left side zero, nonnegativity and continuity force  $\mathcal{D} = 0$  along the orbit, and strictness gives  $F(u_*) = 0$ .

The Lyapunov identity closes the information hidden by projection. If a strictly decreasing quantity returns to its original value, it could not have dissipated anywhere along the orbit; strictness then forces stationarity.

**Uniqueness gate.** If  $K$  is complete,  $T_\tau(K) \subset K$ , and

$$\|T_\tau u - T_\tau v\| \leq q\|u - v\|, \quad 0 \leq q < 1,$$

then  $T_\tau$  has a unique fixed point in  $K$ , and its iterates converge to that point. This gate does not apply without the invariant complete domain.

Banach contraction is correspondingly stronger and more local in its demands. It proves uniqueness only inside the declared complete basin  $K$ . It does not exclude other basins or other physical branches outside that set.

## 9 Schur–Feshbach reduction with domains

Even when complementary modes decay, they can feed back into the coherent sector before disappearing. Schur–Feshbach reduction calculates that feedback. It solves the complementary block in terms of the retained block and substitutes the result back, producing an effective operator rather than simply deleting  $Q$ .

Let  $\mathcal{H} = P\mathcal{H} \oplus Q\mathcal{H}$ . Consider a closed block operator

$$L = \begin{pmatrix} L_{PP} & L_{PQ} \\ L_{QP} & L_{QQ} \end{pmatrix}$$

with domain  $P\mathcal{H} \oplus \mathcal{D}(L_{QQ})$ . Assume:

1.  $0 \in \rho(L_{QQ})$  and  $L_{QQ}^{-1} : Q\mathcal{H} \rightarrow \mathcal{D}(L_{QQ})$  is bounded in the graph norm;
2.  $L_{PP} : P\mathcal{H} \rightarrow P\mathcal{H}$  and  $L_{QP} : P\mathcal{H} \rightarrow Q\mathcal{H}$  are bounded; and
3.  $L_{PQ} : \mathcal{D}(L_{QQ}) \rightarrow P\mathcal{H}$  is graph-norm bounded, so  $L_{PQ}L_{QQ}^{-1}$  is bounded.

**Theorem 9.1** (Schur–Feshbach equation). *Under these hypotheses, solving  $L(p, q) = (f_P, f_Q)$  is equivalent to*

$$Sp = f_P - L_{PQ}L_{QQ}^{-1}f_Q, \quad q = L_{QQ}^{-1}(f_Q - L_{QP}p),$$

where

$$S = L_{PP} - L_{PQ}L_{QQ}^{-1}L_{QP}.$$

If the inverse satisfies the additional graph-norm bound

$$\|L_{QQ}^{-1}\|_{Q\mathcal{H} \rightarrow \mathcal{D}(L_{QQ})_{\text{gr}}} \leq C_Q/\omega_Q,$$

then

$$\|S - L_{PP}\| \leq \|L_{PQ}L_{QQ}^{-1}\| \|L_{QP}\| \leq \frac{C_Q \|L_{PQ}\|_{\text{gr}} \|L_{QP}\|}{\omega_Q}.$$

*Proof.* The  $Q$  block equation gives the displayed formula for  $q$ . Substitution in the  $P$  block equation gives  $S$ . The bound follows from the declared graph norm and inverse estimates.  $\square$

This is a local linear reduction near the reference state. Nonlinear truncation additionally requires control of nonlinear remainders and of the time interval on which the eliminated sector remains small.

The correction  $L_{PQ}L_{QQ}^{-1}L_{QP}$  has a direct interpretation. The retained sector excites a complementary mode through  $L_{QP}$ , that mode propagates through the complementary inverse, and it returns through  $L_{PQ}$ . A large gap can make the inverse small, but only after domains and mixing blocks have been controlled. This is why “the heavy modes decouple” is a conclusion to quantify rather than a slogan.

## 10 Projector stability and basin-local robustness

Two different notions of robustness are needed. Projector stability asks whether the selected spectral subspace itself changes continuously under a perturbation. Basin robustness asks whether a fixed point moves only a controlled amount once two maps act on the same invariant basin. Neither statement follows from the other.

**Proposition 10.1** (Riesz-projector stability). *Let  $A(\epsilon)$  be a norm-resolvent-continuous family on a connected parameter neighborhood, and let a contour  $\Gamma$  remain in the resolvent set while enclosing one isolated cluster. Assume its Riesz projector has finite rank at one parameter value. Then*

$$P(\epsilon) = \frac{1}{2\pi i} \oint_{\Gamma} (z - A(\epsilon))^{-1} dz$$

*is norm continuous and has constant finite rank. This proves projector stability, not dynamical stability of a fixed point.*

The contour in this proposition is the protective spectral boundary. As long as no eigenvalue or spectral cluster crosses it, the rank cannot jump. If the contour ceases to lie in the resolvent set, the selected sector may change dimension and the conclusion no longer applies. Finite rank is a separate hypothesis here: the zero operator on an infinite-dimensional Hilbert space has an isolated spectral point, but its enclosing Riesz projector is the identity and has infinite rank. A physical compact-resolvent realization can supply the finite-rank hypothesis; the abstract Hilbert-bundle notation alone cannot.

**Theorem 10.2** (Basin-local fixed-point robustness). *Let  $T, \tilde{T} : K \rightarrow K$  be contractions on the same complete invariant domain with contraction constant at most  $q < 1$ . If*

$$\sup_{u \in K} \|T(u) - \tilde{T}(u)\| \leq \varepsilon,$$

*then their fixed points satisfy*

$$\|u_* - \tilde{u}_*\| \leq \frac{\varepsilon}{1 - q}.$$

*Proof.* Insert and subtract  $T(\tilde{u}_*)$  and absorb the resulting  $q\|u_* - \tilde{u}_*\|$  term.  $\square$

This is the justified foundation for a local robustness or universality claim. It is not global universality across arbitrary microscopic models.

The factor  $(1 - q)^{-1}$  also shows the cost of weak contraction. When  $q$  is close to one, a small change in the map can move the fixed point substantially. A claim of universality must therefore report both the map discrepancy and the contraction margin.

## 11 Projection, descent, and recovery types

Projection can support an effective theory only if upper states that look the same below continue to look the same after evolution. Otherwise the reduced future depends on hidden information and no autonomous reduced law exists. This is the role of the descent criterion.

Let  $r : X \rightarrow Y_{\text{eff}}$  be a surjective reduction map on a microscopic state space  $X$ , and let  $\Phi : X \rightarrow X$  be a microscopic step.

**Imported autonomous-descent criterion.** The Projection–Admissibility paper is the canonical theorem source [13]. There exists a unique reduced map  $\bar{\Phi} : Y_{\text{eff}} \rightarrow Y_{\text{eff}}$  satisfying

$$\bar{\Phi} \circ r = r \circ \Phi$$

if and only if

$$r(x) = r(x') \implies r(\Phi x) = r(\Phi x').$$

Necessity follows by applying  $\bar{\Phi}$ . For sufficiency, define  $\bar{\Phi}(r(x)) = r(\Phi x)$ ; the implication makes this independent of the representative, and surjectivity gives uniqueness.

A section  $s : Y_{\text{eff}} \rightarrow X$  obeys  $r \circ s = \text{id}$  and chooses one representative. It does not recover every microscopic state. Exact microscopic recovery would require  $s \circ r = \text{id}_X$ , which is possible only when  $r$  is injective. An effective merger combines reduced descriptions and is neither kind of inverse unless separately typed and proved.

The distinction between a section and a decoder is especially important in physical language. A section is a convention for choosing one upper representative of each effective state. A decoder would recover the upper state that actually occurred. When several upper states share one effective label, the first operation may exist while the second is impossible from that label alone.

If  $r$  is fiberwise over the base and the microscopic generator is local in base variables, a reduced local generator may descend when the invariance criterion holds and all coefficients depend locally and smoothly on the base jet. A projection nonlocal in base variables does not inherit locality merely from being a projector.

## 12 Admissibility and hybrid selection

An effective description usually has a domain of trust. Curvature may become too large, a spectral gap may close, a positivity condition may fail, or a truncation error may exceed its tolerance. MTT packages these conditions as admissibility margins so that leaving the domain is an explicit event rather than an unnoticed breakdown of notation.

Let  $m_j(u)$  be declared continuous margins and define

$$\mathcal{A}_\varepsilon = \{u : m_j(u) \geq \varepsilon \text{ for all } j\}, \quad \mathfrak{B}(u) = \max_j [-m_j(u)].$$

Then  $u \in \mathcal{A}_0$  exactly when  $\mathfrak{B}(u) \leq 0$ . This is a diagnostic encoding of declared constraints, not automatically an energy or force.

At a first exit from  $\mathcal{A}_0$ , three logically different constructions are possible:

1. stop the reduced description;
2. continue the upper dynamics and derive a new reduced chart; or
3. impose a reset  $S : \partial\mathcal{A}_0 \rightarrow \mathcal{A}_0$ .

The third option defines a hybrid dynamical system. It changes the continuation law and must separately prove existence, measurability, conservation, and any probability assigned to alternative reset outcomes. Calling the reset “selection” does not make it part of the original flow.

For example, replacing a state by the nearest admissible state after a margin crossing may be a useful numerical prescription. It is nevertheless a new rule. If several replacements are possible, their probabilities cannot be inferred from the existence of the boundary; a measure or dynamical selection mechanism must be supplied.

### 13 Complete admissibility-margin ledger

The ledger below is a practical checklist for constructing or reviewing an MTT realization. It is intentionally longer than a single “closure” condition because each entry can fail independently. A downstream paper should either provide an entry, cite its canonical source, or mark it open.

Every realization must record the following entries independently:

1. **Geometry/domain:** base, fiber, dimensions, bundle regularity, operator domains, and boundary conditions.
2. **Joint spectral structure:** strong commutation or a selected total operator, the spectral contour, rank, and Sobolev boundedness of  $P$ .
3. **Internal gap:**  $\lambda_{\text{int}}$  and its uniformity domain.
4. **Complementary stability:**  $M_Q, \omega_Q$  and the generator sign.
5. **Leakage/invariance:**  $QF(Pu)$  or a quantitative leakage bound.
6. **Existence gate:** the invariant set  $K$  and the compact, condensing, or contraction theorem actually used.
7. **Equilibrium gate:** stationarity equation or strict Lyapunov promotion.
8. **Coherent contraction:** norm, basin, and factor  $q < 1$ .
9. **Truncation:** mixing blocks, inverse/domain bounds, nonlinear remainder, and error interval.
10. **Projector robustness:** perturbation topology and resolvent contour.
11. **Admissibility:** each margin  $m_j$ , its units, and its source.
12. **Continuation/selection:** stop, upper continuation, or hybrid reset, including conservation and probability data.
13. **Physical hyperbolicity:** principal symbol, constraints, causal domain, and relation between  $t$  and  $\tau$ .
14. **Scale separation:** record the internal gap  $\lambda_{\text{int}}$ , coherent contraction scale, four-dimensional cutoff, curvature scale, and RG scale  $\mu$ .
15. **Numerical provenance:** source-independent inputs, fitted inputs, branch selection, uncertainty, and held-out outputs.

No single scalar “coherence scale” may replace this ledger unless a theorem derives all identifications with dimensions and errors.

The ledger also prevents parameters with different units from being merged. An internal spectral gap, a contraction rate, a four-dimensional cutoff, and an RG scale can all be numerically expressed as energies after conventions are chosen, but equality of their numerical

values would still require a same-source theorem.

## 14 Lorentzian physical completion

The abstract Hilbert structure is designed for control estimates and carries a positive norm. Physical spacetime needs a different piece of information: a hyperbolic equation whose principal symbol distinguishes one temporal direction from spatial directions. Trying to obtain Lorentzian signature by changing the sign of a positive Gram tensor confuses these roles.

The Hilbert inner product and every Gram tensor constructed from it are positive semidefinite. They cannot acquire Lorentzian signature. In a local physical completion, causal signature is instead read from the principal symbol. For a second-order field equation this has the schematic form

$$\sigma_{\text{pr}}(x, \xi) = g^{\mu\nu}(x) \xi_\mu \xi_\nu I + \text{gauge blocks}.$$

Lorentzian hyperbolicity, constraint propagation, and a domain-of-dependence theorem must be verified for the selected equations. The canonical base dimension  $3 + 1$  is part of the canonical FP physical realization; it is not derived by the abstract Hilbert-bundle theory. In particular, the component identity  $1 + 3 \times 3 = 4 + 6$  neither selects a four-dimensional causal base nor determines its signature.

In other words, the Foundation can prove that a selected internal sector is stable while remaining neutral about which physical equation propagates it. The signature paper and any later action supply that equation. This separation allows the same control spine to be used with different physical completions, but it also means that causality is not derived from projection alone.

## 15 Foundation synthesis theorem

### What the later finite calculus supplies

The Cohesive Closure Repair owner gives a useful downstream example, not a replacement for the analytic hypotheses [2, 7, 8, 9, 10]. Its exact 144-dimensional symmetric Weyl DGA retracts to a 48-dimensional complex retaining the old 36-dimensional response and twelve higher-jet harmonic classes. This is not a minimal model: the differential is nonzero. Nonzero  $m_3$  and  $m_4$  accompany the nonassociative binary product, and a proved recurrence gives a nonzero operation at every higher arity on a selected input family. That is not a complete operation or residual table from  $m_5$  onward. Exact contraction-preserving DGA maps transport all arities by the tree formula; approximate maps require defect and tail bounds.

For a physical application, these identities still require a selected operator on a declared domain and the relevant invariant basin, derivative, gap and resolvent estimates. A finite transfer operation is not automatically a physical interaction, and a positive repair cost is not the signed physical action. The independent Fixed Points gates above remain unchanged.

The owner's endpoint factorization makes the physical input ledger more precise [11]. Its seven operator/action/compactification rows depend on three same-source types: geometry-action data (GAS), spectral synthesis with domains and error control (SYN), and four-dimensional BV-compatible compactification (BV4). These are structured objects, not three scalar parameters. The action Hessian and synthesis determine the Galerkin residual and,

when the complementary inverse exists, its Feshbach correction; that execution introduces no fourth source type. None of the three physical packets is supplied by the abstract Foundation. Established hidden projective existence and finite/profile results are retained; the selected common-chamber endpoint, literal metrics, physical action and compactification remain the concrete boundary.

The synthesis theorem gathers the preceding gates without weakening their independence. It should be read as a conditional assembly instruction: when a realization supplies each named hypothesis, the corresponding control conclusion follows. It is not a shortcut by which the word “MTT” supplies all hypotheses at once.

**Theorem 15.1** (Scoped MTT foundation). *Assume the dimension-neutral Hilbert-bundle architecture, joint spectral structure, internal gap, stable complementary semigroup, and the separately listed hypotheses for a chosen fixed-point theorem. Then the model possesses a bounded coherent decomposition and a projected time-step fixed point. Under strict Lyapunov and invariance hypotheses that point is an equilibrium. Under the Banach gate it is unique in the declared basin. Under the Schur–Feshbach hypotheses the local linear complementary sector can be eliminated with the displayed error bound. Under contraction and map-closeness hypotheses the fixed point is basin-locally robust. Autonomous reduced dynamics exists exactly under the descent criterion.*

None of these conclusions supplies a physical probability law, Lorentzian field equation, quantum representation, particle spectrum, Standard Model matching, cosmology, or numerical prediction. Those are downstream obligations governed by the admissibility ledger. Nor does equality of the local  $1 + 2 + 3$  strain dimensions and the q79  $1 + 2 + 3$  trace-split ranks prove the same-source intertwiner needed to identify their metrics, connections, and vertical operators. Theorem 3.1 closes the common flat line and finite Hessian square inside the q79 carrier; it does not close that local-to-continuum identification.

What a downstream paper may legitimately inherit is now explicit: a coherent spectral decomposition, conditional fixed-point and equilibrium gates, controlled linear elimination, local robustness, and an exact finite shared-line square, each on its declared hypotheses. What it may not inherit without new work is a physical state space, a gauge or particle representation, a value source, or a causal field equation. This is the operational meaning of calling the present paper a foundation.

## 16 Conclusion

The corrected Foundation provides a typed and noncircular spine for the MTT corpus. It identifies what the triplet and projector mean, how complementary stability is proved, which fixed-point conclusion is available, how reduction is typed, and where physical assumptions enter. Its value is precisely this separation: later realizations can now be tested against explicit gates rather than inheriting conclusions from an undifferentiated appeal to coherence. The shared-circle sector now illustrates the intended discipline particularly well: a curved Boothby–Wang schema, an exact flat q79 differential line, and a conditional polarized Hilbert readout coexist without being conflated.

The broader research direction is to replace the remaining compatibilities by commuting source maps. The highest-value next step is not another rank count or a restatement of the fixed-point gates. It is a same-source continuum comparison using the nonlinear strain shadow, the physical q79 HYM complex, its connection, shorted Hessian and normalized response data. An excluded linear root-space embedding is not that comparison. With the

required maps and a selected action, the abstract control spine could begin to explain why the successful lower theories arise from one source rather than merely fit together after selection.

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## Computational Evidence and Reproducibility

The newly imported results are cited individually in the bibliography at the immutable curated source state, with local result identifiers recorded in the metadata companion. The archive is <https://github.com/PeterNero/mtt-results-repro>. Exact finite calculus, conditional continuum implications and physical source obligations retain their distinct tiers; an archive citation is not an independent proof check.